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Unraveling the Complexity: Empowering Decision-Making through a Multi-Agent Decision Support System

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Abstract- Deciphering the intricate landscape of financial markets requires astute decision-making based on a myriad of factors. In this groundbreaking research, I introduce a pioneering solution: a cutting-edge decision support system tailored to empower users in their stock selection endeavors. Through the fusion of fuzzy inference systems, genetic programming, and a zero-sum multi-agent architecture, I revolutionize the way financial decisions are made. Within this innovative framework, each agent harnesses the power of fuzzy inference systems to make informed determinations regarding trading direction and optimal trading force. By unraveling the evolved membership functions embedded within the system, market participants gain profound insights into the dynamic nature of finance. Leveraging the power of regression analysis on stock time series data, our multi-agent system accurately predicts the future movements and strengths of various financial markets. Rigorous testing of our methodology showcases its prowess, as it outperforms a benchmark based on the illustrious Dow Jones Index. Armed with this comprehensive tool, investors are equipped to navigate the ever-shifting tides of the financial realm with confidence. Our research not only pioneers a new era in decision support systems but also unveils an indispensable resource that empowers individuals to make strategic, finance-savvy choices in their stock trading endeavors.

I. Introduction

In this paper, I will introduce a method that uses Agent-Based Modeling (ABM, [1]) to improve the decision process used in trading financial markets: the trader will receive a recommendation of what financial market to trade, as in the work of Brown, Pelosi, and Dirska [2]. The proposed method uses a combination of fuzzy inference systems (FIS), genetic programming (GP) [3] [4] and multi-agent systems (MAS) [5] to generate models that can explain the behavior of complex systems such as that of a financial market by analyzing the evolved membership functions (MF) of each agent. The rationale behind this selection of approaches is as follows: GP provides more flexibility against a more traditional approach, such as parameter tuning; MAS represents a model that simulates how a the market is actually generated through the interaction of several entities that are buying and selling that market, and an FIS is a model that can provide knowledge representation through a set of rules and MFs that can be interpreted by a human being, unlike other models that act as a black box.

The MAS developed in this work belongs to a class of coalitional games called constant-sum game [5], and, in particular, it is a zero-sum game. Agents in the proposed method are selfish agents that don't intervene with other agents, but there is a higher-order process that is dictating how they must organize. In this case, the collective work of every agent must yield a "zero-sum": the sum of their forces must be equal to the observed price, or in other words, they must adjust to the real prices. To achieve this, each agent uses knowledge represented as an FIS to decide how much trading force it is going to provide and in what direction, and its MFs are generated using GP. The combination of using FIS and GP allow us to create conclusions about the behavior of the market by interpreting the evolved MF of each agent in the system (for example, one can infer that whenever a certain TI is low and another TI is high, the market will stagnate). Specifically, the predicates and consequents of the FIS are generated using the sum of sines that are found using a modified algorithm based on GP. The sum of sines was used because, in theory, they are capable of fitting any curve. In the end, the developed algorithm gives, as a result, the community of agents that obtained the absolute value of the lowest mean-squared error (MSE) plus the sum of its agents scores (explained in Section III).

The rest of the paper is structured as follows: next, I will present an overview of the state of the art in the use of multi-agent systems for financial forecasting, as well as other soft computing techniques, followed by a presentation of the methodology followed in III whose results are presented in Section IV. Finally, I will draw our conclusions and discuss future lines of work.

II. Related Work

The core algorithm of the proposed method is, at its highest level, a multi-agent system. It is, therefore, paramount to mention some related works that use MAS

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to forecast or understand financial markets.

Klingert and Meyer [6] implement a MAS to analyze the effect of two market mechanisms: the continuous double auction and logarithmic market scoring rule. The purpose of the agent-based simulation model is to see the impact on the number of trades, the accuracy of prediction markets and the standard deviation of the prices in order to prove the three hypotheses that they propose. In the end, due to a higher amount of trades and lower standard deviation of the price, their results indicate that the logarithmic market scoring rule seems to have an advantage over the other mechanism.

Sherstov and Stone [7] present three automated stock-trading agents that follow different strategies to predict financial markets and are compared. The first agent uses reinforcement learning, the second a trend-following strategy, and the last one is marketmaking. These agents are part of a MAS where the better-performing agent is chosen for the testing phase.

Kendall and Su [8] use a MAS to simulate stock markets within which stock traders are modeled as heterogeneous adaptive artificial agents. On average, 80% of the artificial stock traders were able to trade using successful trading strategies, which brings the investors have higher returns compared to a simple buy-and-hold strategy.

The authors of this work gained useful knowledge about MAS from two theses. The first one is the work from Grothmann [9]. This work served as inspiration for the architecture of the proposed method. The second thesis is by Boer-Sorbn [10], which gave an overview of approaches to describe and understand the financial market's dynamics and motivated the authors of this work to use the approach of agent-based computation to perform financial forecasts.

GP, the method used in this paper to generate the membership functions has been previously used in many fields of economics, generally to create regression models or as a means finding a better architecture for trading strategies. In the the proposed method, GP is used to generate the MF of the FIS that serve as the agents' functions in the MAS, an uncommon technique; however, It is more common to find GP used to generate trading rules. For instance, Li and Tsang [11] developed a system that generates decision trees of TI using GP. Preliminary results showed that it outperforms commonly used, non-adaptive, individual technical rules with respect to prediction accuracy and average annualized rate of return over two different out-of-sample test periods (three and a half year in each period).

Garcia-Almanza and Tsang [12] used GP as a regression tool and a technique called repository method to model "rare instances" or emergent phenomena. The repository method is a technique that analyses decision trees produced by GP to discover classification rules. It lets us model the rare instances in different ways, increasing the possibility of identifying similar cases in the future in the time series. Garcia-Almanza's work was useful for the authors of this work, as it noted the importance of determining emergent phenomena in financial markets and stressed the usefulness of ABM in modeling them.

An artificial market that models technical, fundamental, and noise traders were developed by Martinez-Jaramillo and Tsang [13]. In this work, GP is used to generate the agent functions [14] of the technical traders, and TI were used as the operators for the GP algorithm.

Closest to us is the work by Chen and Yeh [15], which propose an architecture based on GP and ABM that takes trader's (agents) Search behavior densities, and by using simulated annealing, connect these behaviors to psychological factors, such as peer pressure or economic factors such as the standard of living.

As has been previously mentioned, each agent uses an FIS in this work. The agent function has the objective of determining how much "trading force" an agent has to provide, so the sum of all the agent forces gives a zero-sum system. FIS has been used to predict financial markets successfully. This subsection provides a series of works that use fuzzy logic and other techniques related to the proposed method as a means to forecast financial markets. Ijegwa et al. [16], created a FIS that uses four TI as input, and the output is a recommendation to buy, sell or hold in a financial market. This way, their system aids the trader in the decisionmaking process. They decided to use the moving average convergence/divergence, relative strength index, stochastic oscillator, and on-balance volume as the TI inputs for the FIS.

Huang, Pasquier, and Quek [17] describe the application of a hierarchical coevolutionary fuzzy system called HiCEFS for predicting financial time series. Their system is based on the TI price percentage oscillator. An interesting part of this work is their use of irregularly shaped membership functions (ISMF). The employment of ISMF allows their system to construct an accurate predictive model, and it outperforms the simple buy-and-hold strategy.

III. METHODOLOGY

First, I had to settle on a dataset on which I could perform this work. Initially, I selected the dataset proposed by Brown, Pelosi, and Dirska [2], which can be found at the UCI Machine Learning Repository is under the name "Dow Jones Index Data Set." This dataset was selected because it contains recent trading data and it is available in an open repository.

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In this work, I only used the open, high, low, and close prices, and we obtained the average directional index (ADX), stochastic oscillator (SO), and relative strength index (RSI) for the dataset records by using an online tool called FreeStockCharts, which can be found at http://www.freestockcharts.com/. I added the TI to the stocks provided in the dataset by Brown et al. using FreeStockCharts and downloaded the full historical data from the platform. Filtering was performed where only the relevant records were extracted (the same records that are present in the benchmarking dataset).

This dataset was enriched with another attribute, which I call "directional strength" (DS), which indicates how strong a movement was at a particular moment in a time series, and What was its direction (downtrend or uptrend)? An absolute value of 100 of the directional strength means that it was a very strong movement, where all the traders in that financial market agree to buy or sell, and a value of 0 means strong indecision, a low change in the value, with a big difference between the opening and closing prices. The formula to calculate the directional strength, which I introduce for the first time in this paper, is as follows:

$$DS(t) = 100\frac{C(t) - O(t)}{H(t) - L(t)} \qquad (1)$$

Where t represents a particular point of time in a time series, O represents the open price, H the high, L the low, and C the close price. This dataset is going to be used to create a system that helps make decisions regarding stock trading, by telling the user the the direction the market is going to take (downtrend or uptrend), and its strength, through the use of the directional strength indicator. The system I introduce in this paper is composed of a community of agents, which are the final actuators in the environment and simulate the market actors. The action of every agent in the community involves providing certain trading force, which is a value between -100 and 100. The sum of all of the forces of the agents in a community must be equal to the observed or real directional strength indicator for a particular record in the dataset. A community receives its input, which are the ADX(t), SO(t), and RSI(t) indicators, and outputs DS(t + 1), the directional strength of a future record, i.e., one data point in the future.

The method creates several communities, which exchange genetic material of its agents among them in order to produce better-performing communities, and their fitness is determined using a sum of the mean-squared error (MSE) and the absolute value of the sum of the forces of its agents. The use of this summation was determined to be a good fitness indicator by trial and error, but other fitness functions can be used. A flowchart depicting the algorithm that creates a population of these communities is presented in Figure 1.

Each of the agents in the communities uses a FIS to make decisions. The inputs of this FIS, as has been mentioned before, are the ADX, SO, and RSI.

The output is a real number between -100 and 100 that represents the directional strength. To arrive at this output number, one can choose a number of MFs that act as inputs and a number of MFs that act as outputs. By trial and error, the authors found that a number of five membership functions gave good results in most cases. As the MF are being evolved, they don't really mean something such as "high" or "low," as in other fuzzy systems constructed by expert human beings. The objective of this process is only to map an antecedent to a consequent, to later perform an aggregation and defuzzification.

The FIS demulsifies the aggregation of the MF at their activation level determined by its inputs, and this will represent a directional strength for that particular agent. The rule set is depicted below and is comprised of 15 rules: 5 outputs for each of the 3 TI. GPpredN represents a predicate MF created by the GP algorithm and, similarly, GPconsN represents a consequent.

1) if ADX is GPpred1 then DS is GPcons1
2) if ADX is GPpred2 then DS is GPcons2
3) if ADX is GPpred3 then DS is GPcons3
4) if ADX is GPpred4 then DS is GPcons4
5) if ADX is GPpred5 then DS is GPcons5
6) if RSI is GPpred6 then DS is GPcons6
7) if RSI is GPpred7 then DS is GPcons7
8) if RSI is GPpred8 then DS is GPcons8
9) if RSI is GPpred9 then DS is GPcons9
10) if RSI is GPpred10 then DS is GPcons10
11) if SO is GPpred11 then DS is GPcons11
12) if SO is GPpred12 then DS is GPcons12
13) if SO is GPpred13 then DS is GPcons13
14) if SO is GPpred14 then DS is GPcons14
15) if SO is GPpred15 then DS is GPcons15

In order to create the MF, a GP algorithm performs a number of operations among the initially generated communities. These operations are the classical crossover and mutation operations, along with two new operations: migration and replacement. The crossover operation chooses a subtree of the GP function of the ith agent of a community A and passes it to the GP function of the ith agent of a Community B. A subtree from the community B is passed to the GP function of the ith agent. As a result, two new agents are generated, and the old agents are replaced. To determine what communities are to be crossed over, a tournament is held where two random communities are chosen from the population, and the one with the lowest MSE plus the absolute value of the sum of forces (as explained above) is chosen; this will obtain the first community. This process is repeated to obtain a second community. As for the mutation operation, a

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subtree from community A is selected and randomly changed to produce a new GP function. The community to be mutated is randomly selected from the population.

The migration operation is rather simple. According to the migration rate, an agent from community A will be interchanged with another agent from community B. In order to determine if this interchange is to be performed, the algorithm checks if the sum of the forces of community A is of a different sign than the sum of forces of community B. This ensures that the migration will produce a community closer to a zero-sum, i.e., the community will be closer to the observed or real prices. As for the replacement operation, a whole new community is generated and replaces the community with the highest absolute value of the MSE plus the sum of forces. All four operations have a pre-established chance of occurring (except for the replace chance, which is currently being dynamically adapted by using a FIS. One can see a flowchart of the general process in Figure 1.

In addition to this set of operations, the GP algorithm uses the parameters of population size (in this case, the number of communities), the number of agents per community, the number of generations, how many inputs the agents will process (in this case, the TI), and how many outputs (which, in the end, as has been noted previously, get defuzzified into a directional strength value).

In this paper, I introduce a method that generates the MF: The GP algorithm uses a sum of sines to produce functions that are converted to the MF of the agents' FIS. A graphical representation of an input and output MF is depicted in Figure 3. The GP algorithm uses only two operators: the sum and the sine operators. The sum of sines is represented by

$$y = \sum_{i=1}^{n} a_i sin(b_i x + c_i) \qquad (2)$$

Taking into consideration Formula 2, the GP algorithm uses the following literals set: a, b, and c are random real numbers, where a ranges from 0.0 to 0.3, b ranges from 0.0 to 10.0, and c ranges from 0.0 to 1.0. These ranges were obtained by trial and error, and other ranges could perform better. An additional real number is added to the set, which ranges from -1 to 1, along with the variable x, which takes the value of the current input (ADX, RSI or SO).

The replacement rate, instead of being constant, is dynamically adapted using a FIS while the GP process is running. This idea was adapted from the series of works by Castillo et al. An example of this dynamic adaptation of parameters can be found in [18], where the authors apply a FIS that controls the value of certain parameters in an ant colony optimization algorithm (ACO), proposing different architectures, which are then compared in performance. In the end, the authors prove that the dynamic adaptation of parameters through the use of FIS can improve the performance of ACO. Additionally, one can find a survey of the application of this type of technique in the work by Valdez, Melin and Castillo [19].

In the proposed method, a similar approach is taken, where dynamic adaptation of parameters through FIS is used to avoid stagnation of the GP algorithm. In this case, a stagnated GP algorithm would mean that it has been producing the same fitness over a number of generations, i.e., the absolute value of MSE plus the sum of forces is not being lowered in the last N generations. To calculate this stagnation level, the authors of this work propose the stagnation index, which is described by Formula 3, where S(P) means the stagnation index of P, where P is the set of fitnesses over a certain number of generations and C(P) is the count of repeated fitnesses over the last N generations. For example, given the set P = 5, 5, 5, 3, 2, 1, C(P) = 3.

$$S(P) = 1 - \sqrt{\frac{1}{C(P)}} \qquad (3)$$

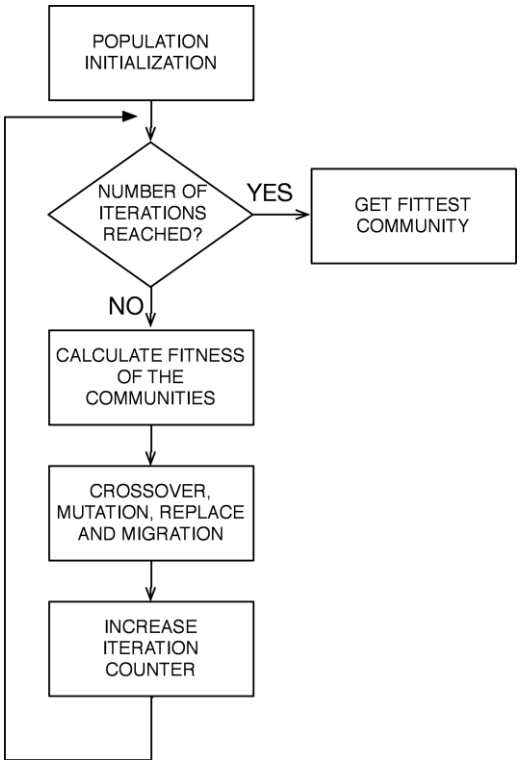

Figure 1. Flowchart of the Proposed Method with Constant Replace Chance

The stagnation index is given as input to a FIS, which gives as output a new replace chance for the replace operation. The objective of this dynamic adaptation of parameters is to obtain lower errors in a smaller amount of generations. This is proven in Section IV. Also, it is worth mentioning that this technique could be applied to

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the crossover, migration, and mutation chances. Still, due to a limitation of time, the authors of this work could only apply the technique to one parameter. A flowchart depicting the modified GP algorithm is shown in Figure 4.

IV. Experiments And Results

This part starts by making sense of the analysis and results got from carrying out the powerful variation of the substitution opportunity through the utilization of a FIS. Then, the proposed technique is analyzed against the strategy proposed by Brown, Pelosi, and Dirska [2], utilizing their proposed dataset. The analysis by Brown et al. was extended to 30 preliminaries rather than 8 to acquire more factual proof. Besides, a progression of investigations (where a winning stock must be picked, as in the work by Brown et al.) with a rising number of ages were performed and, once more, 30 preliminaries were run rather than 8, with 30 different number of ages. One more arrangement of investigations like the examinations referenced before were performed, where various setups of the proposed technique are utilized. Every one of the various arrangements was run for 30 preliminaries, and the typical return for capital invested was gathered.

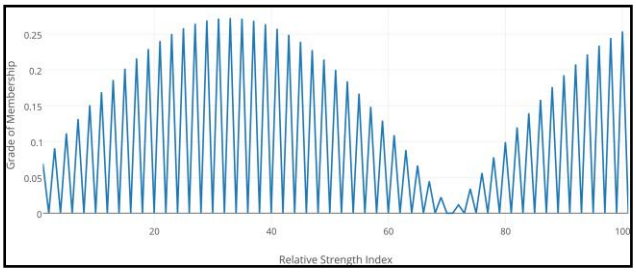

Figure 2. Graphical Representation of an Input Membership Function

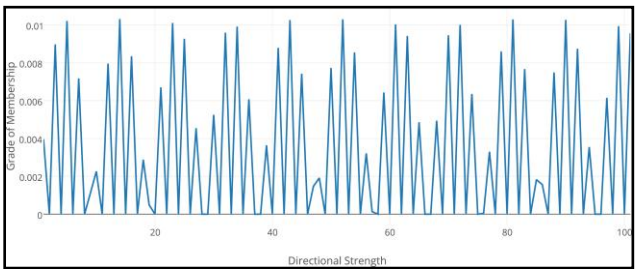

Figure 3. Graphical Representation of an Output Membership Function

After the past arrangement of investigations were played out, the creators chosen to perform tests where the gauge of individual stocks occurred as opposed to picking a stock to purchase for every one of the weeks in the dataset. This arrangement of investigations is more a relapse sort of explore than a suggestion type. Initial, five stocks were picked, and the proposed strategy was applied to various designs. Then, every one of the stocks in the dataset was assessed utilizing the proposed technique and a design that was found to have a decent presentation by experimentation. The adequacy of the not entirely set in stone by how much normal return for money invested is gathered.

As a last examination, the proposed strategy plays out an unadulterated relapse task, where it attempts to anticipate the directional strength of the following week's record in the dataset.

The plan of the FIS is fairly straightforward. The contribution of the FIS is the stagnation file and the result is the new swap rate for the GP framework in that specific emphasis or age. The information MF are portrayed in Figure ?? and the result MF are portrayed in Figure ??. The plan of these MFs were gotten by the emotional assessment of the creators, and by experimentation. The standard set is introduced underneath:

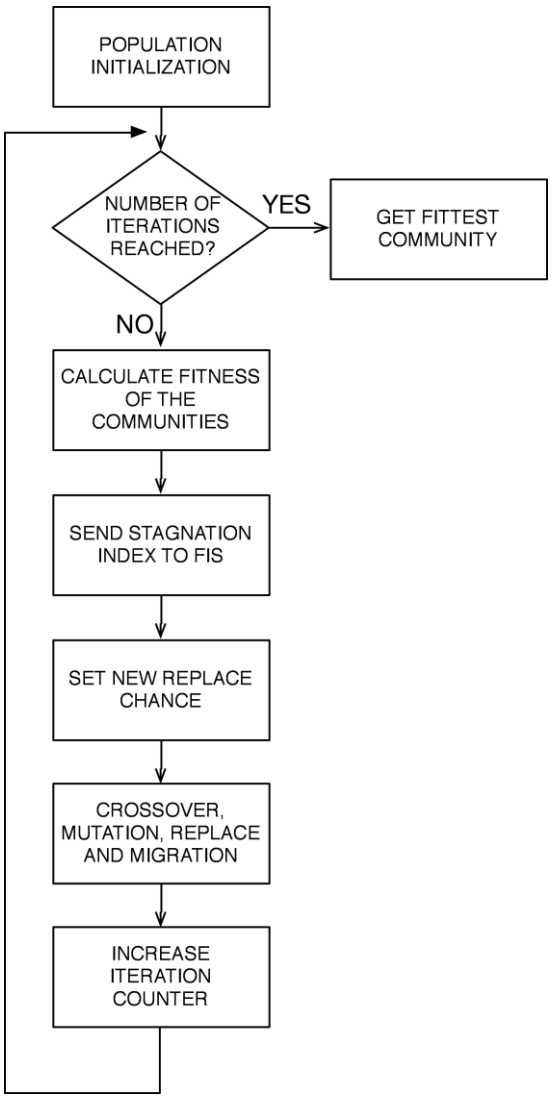

Figure 4. Flowchart of the proposed method with a dynamic adaptation of the replacement rate

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1) if SI is Low then RC is Low
2) if SI is Medium then RC is Medium
3) if SI is High then RC is High

In Figure ??, one can perceive how the dynamic transformation of the substitution rate stays away from stagnation of the proposed technique. By the by, a speculation test was performed to officially demonstrate this speculation. The speculation test was acted in the measurements programming Stata, and the outcomes are displayed in Figure ??. One can see that the speculation test gotten a t-score of - 3.3529, where the basic t an incentive for a 95% certainty stretch is - 2.0018. As an end, the invalid speculation is dismissed (which is that the normal of the blunder of the 30 midpoints of the tests with a unique transformation of the supplant opportunity is more noteworthy than or equivalent to the midpoints without the dynamic transformation), and this implies that the proposed technique a dynamic transformation of the supplant opportunity beats the proposed technique with a consistent worth of substitution rate.

A. Stock Selection

In the accompanying tests, the dataset gave in [2] is utilized, and the primary quarter (12 records) of each stock is utilized for preparing with the subsequent quarter (13 records) utilized for testing. 1) Comparison Against Brown et al.: Brown et al. [2] used their strategy to run a set of 8 fundamentals, where the inspiration driving the assessment is to pick what the estimation considers will be the betterperforming stock to buy for the multi week from now. At last, a lone starter adds up to the speed of return from buying all of the stocks that were picked by the computation. In their work, they present a table appearance what stocks were picked and what the hard and fast speed of return was for their best starter (which, for their circumstance, was the fourth fundamental). Then, at that point, they show the delayed consequences of all the starters in a similar framework to Figure 9, and they give a typical of the eight fundamentals.

The makers of this work don't trust this examination to be adequate to show the feasibility of the proposed technique. Regardless, I will give a preliminary like theirs to check the results out. Table ?? shows what stocks our proposed procedure picked, close by the outright speed of return. For this present circumstance, it defeated the results gained by Brown et al., which was 9.51%.

As referenced previously, Figure 9 shows a graphical portrayal of the outcomes got from the eight preliminaries, where the quarterly return and the typical week by week return are shown. In this case, the typical pace of return of the eight preliminaries was 8.1140%, which again beats the typical got by Brown et al., which was 7.075%. The calculation makes a stock choice, and that intends that it will figure out what stock to purchase at one point in time. The chosen stock is purchased and held, and after a specific period, a benefit or misfortune can be determined relying upon how much focuses the market moved. These focuses can be meant a pace of return.

Table I : Stocks Selection for Trial 4

Week	Stock	Return
1	MMM	0.58265
2	HD	1.92256
3	AA	3.72861
4	CSCO	3.48494
5	CSCO	0.285551
6	HD	0.216626
7	HD	0.981194
8	MMM	1.27858
9	IBM	-2.01259
10	INTC	-1.92661
11	PFE	0.896414
12	AA	3.81731
13	PFE	3.23383
Total		16.489065

2) Extended Experiments - Constant Configuration, Increasing Generations: The essential widened attempt includes giving the proposed strategy with a particular plan (4 organizations, four subject matter experts, three wellsprings of data, and 5 yields), which were heuristically shown to yield extraordinary results. This course of action was run on different occasions for 30 exceptional potential gains of ages. The results are the common speeds of return of the 30 starters, and are depicted in Figure ??. This chart shows that the proposed method will overall get under-ready with a low number of ages, begins to throw extraordinary results with a moderate number of ages, and starts to move past ready with a bigger number of ages. With a very huge number of ages, the system starts acting violently. It is crucial to determine that the proposed system never cultivated a negative ordinary speed of return. The inspiration driving these tests is to show that the calculation acts as other learning calculations normally act: a preparation interaction including a low number of emphasess or ages makes an under-prepared model, a moderate number of emphasess makes a model that performs indeed, and countless cycles makes an over-prepared model.

Table II
CHOOSING WHAT STOCK TO BUY - VARIABLE CONFIGURATIONS

	NUMBER OF AGENTS				
GEN.	1	2	5	10	20
2	1.27%	2.46%	2.33%	1.68%	2.44%
5	3.22%	2.57%	0.97%	2.90%	3.70%
10	1.82%	1.75%	3.04%	3.19%	3.14%
20	2.32%	1.77%	2.28%	1.76%	4.82%
40	3.35%	4.07%	0.58%	4.13%	2.53%
80	1.48%	0.85%	3.34%	5.61%	2.81%

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3) Extended Experiments - Variable Configurations: This trial is like that introduced by Brown, et al., with the distinction that 30 preliminaries are run rather than 8, and unique arrangements of the proposed technique are utilized. The outcomes are shown in Table 4.

A heatmap is provided in Figure 11 for the reader in order to appreciate better the results obtained.

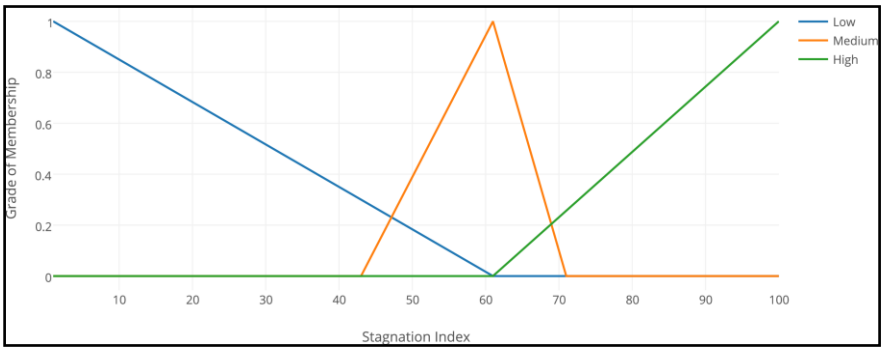

Figure 5. Input for the Dynamic Adaptation of the Replace Chance

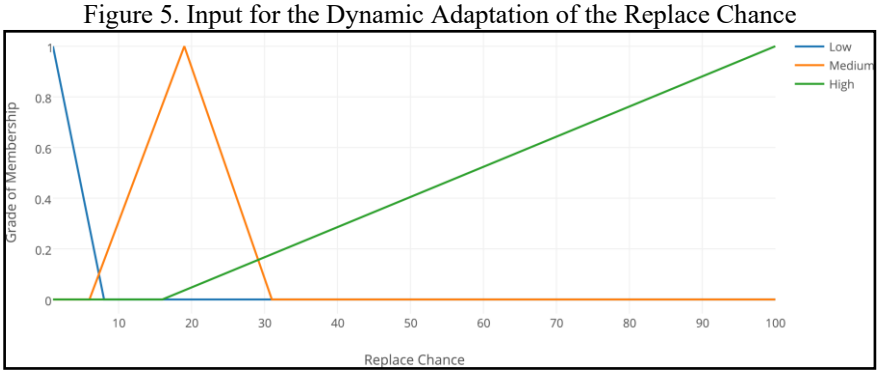

Figure 6. Output for the Dynamic Adaptation of the Replace Chance

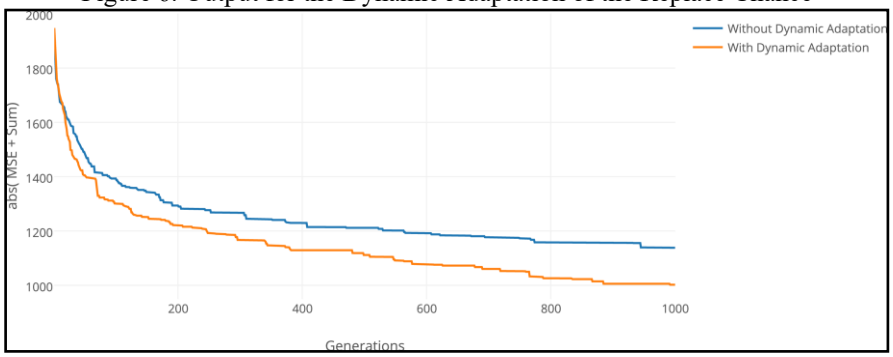

Figure 7. Dynamic Adaptation of Replace Chance vs Constant Replace Chance

Two-sample t test with equal variances

Variable	Obs	Mean	Std. Err.	Std. Dev.	[99% Conf.	Interval]
with_fis	30	1142.388	20.95996	114.8024	1084.614	1200.162
withou~s	30	1244.283	22.00501	120.5264	1183.629	1304.937
combined	60	1193.335	16.46105	127.5067	1149.52	1237.151
diff		-101.8954	30.38981		-182.8322	-20.95866

```
    diff = mean(with_fis) - mean(without_fis)                     t =  -3.3529
Ho: diff = 0                                     degrees of freedom =       58

    Ha: diff < 0                 Ha: diff != 0                 Ha: diff > 0
 Pr(T < t) = 0.0007         Pr(|T| > |t|) = 0.0014          Pr(T > t) = 0.9993
```

Figure 8. Hypothesis Test for the Dynamic Adaptation of the Replace Chance

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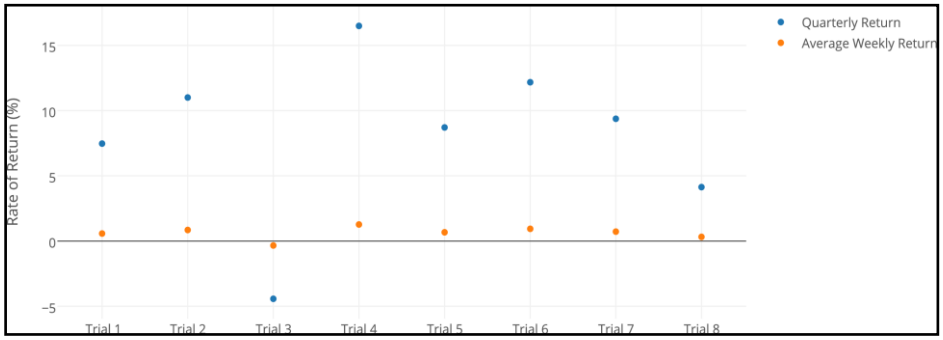

Figure 9. Rate of Return of the 8 Trials

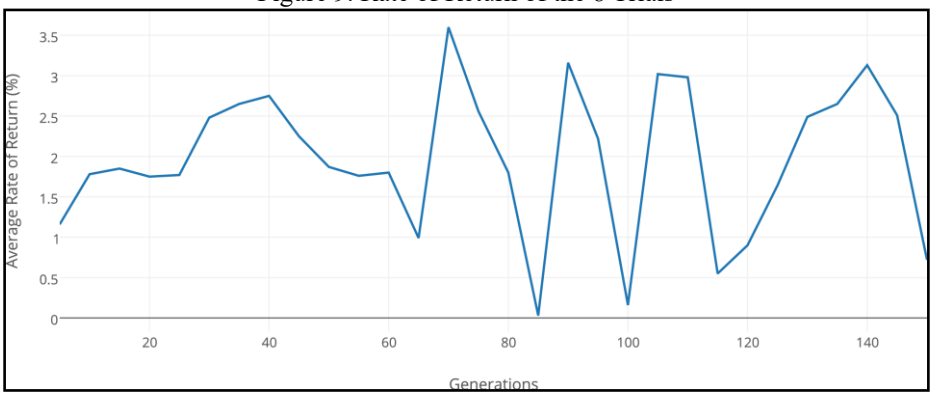

Figure 10. Experiments of the Proposed Method with a Constant Configuration and Increasing Generations

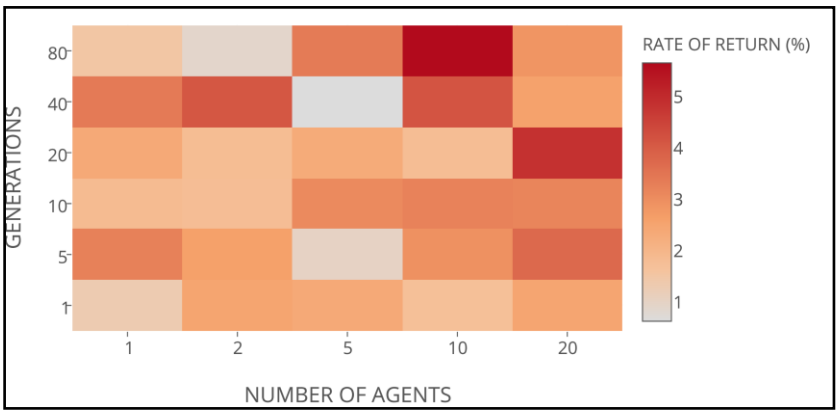

Figure 11. What stock to choose

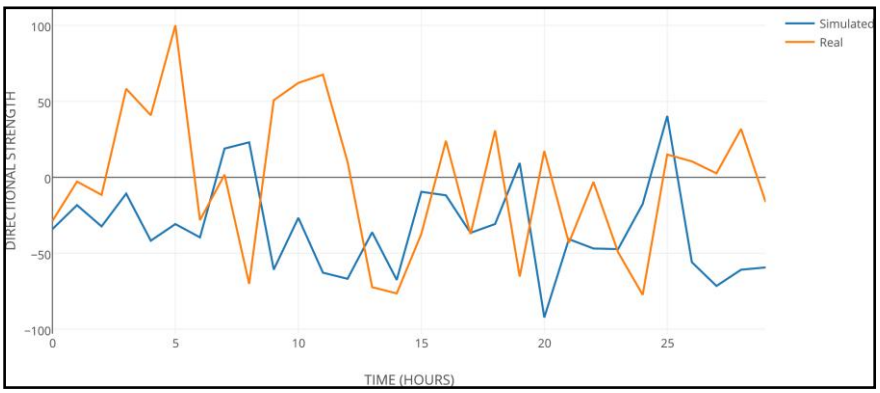

Figure 12. Directional Strength Simulation - 100 Agents, 10 Generations

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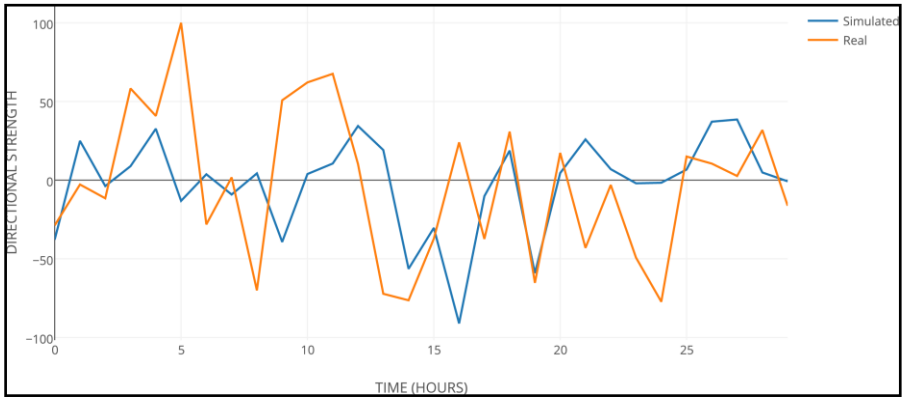

Figure 13. Directional Strength Simulation - 100 Agents, 100 Generations

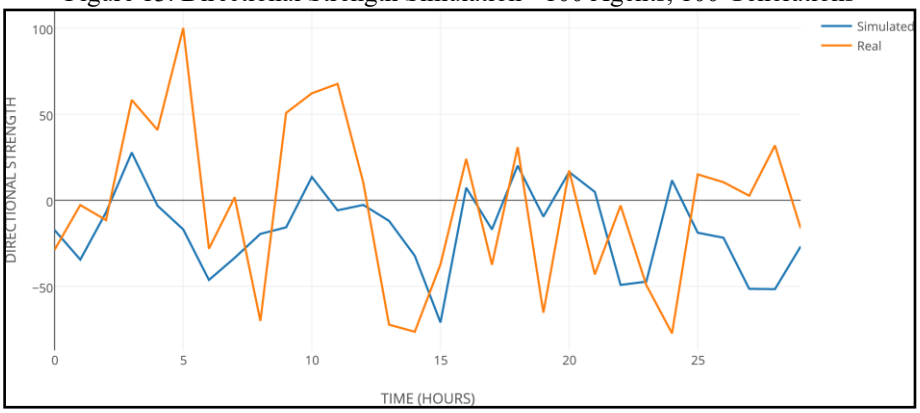

Figure 14. Directional Strength Simulation - 100 Agents, 1000 Generations

V. Conclusion And Future Work

The objective of the present work is to provide a novel method to perform market selection and forecasting. The proposed method achieved this objective by creating a solution based on several algorithms: the use of a MAS to simulate the interaction of traders buying and selling a market, an FIS for each of the agents of the MAS to represent their knowledge of the market, and GP will evolve this knowledge.

The results obtained by using the proposed method are better compared to the work of Brown, Pelosi, and Dirska [2], and show that its use would yield profits if trading the stock market using a stock selection strategy.

An interesting feature of the present method is its use of GP for evolving membership functions. Limiting the possible operators to use in the evolutionary process to the addition and sine operators allowed GP to create flexible tasks that were able to represent the agents' knowledge in the MAS. It is worth noting that the proposed algorithm goes deeper than just giving recommendations; in the end, a novel method for performing regression has been created, and the authors believe that it can be extended also to perform classification tasks.

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